KEVIN AMPONG OCRAN

Aspiring Data Scientist | Data Science Student

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EDUCATION

Bachelor of Science in Data Science and Analytics

Ghana Communication Technology University (GCTU) • Expected Graduation: 2028

Current Focus: Statistical methods, machine learning algorithms, database management, and data visualization

PROFESSIONAL SUMMARY

Data science student with strong foundational skills in Python, SQL, and statistical analysis developed through academic coursework and self-directed learning. Completed intensive 3-month Data Science & Analytics training at Thrive Africa covering machine learning, data cleaning, and visualization. Built 3 end-to-end projects demonstrating ability to analyze datasets, engineer features, train classification models, and communicate results. Seeking data science internship to apply classroom knowledge in real-world business settings and continue developing professional skills under experienced mentorship.

TECHNICAL SKILLS

- **Programming & Tools:** Python, SQL (PostgreSQL), Git/GitHub, Jupyter Notebook, VS Code, Google Colab
- **Data Analysis:** Pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Statistical Analysis
- **Machine Learning:** Scikit-learn (Logistic Regression, Random Forest, XGBoost), Model Evaluation (Accuracy, Precision, Recall, F1-Score, ROC-AUC), Hyperparameter Tuning (RandomizedSearchCV)
- **Data Visualization:** Matplotlib, Seaborn, Plotly, Statistical Plots (Confusion Matrices, ROC Curves, Correlation Heatmaps)
- **Database & Data Engineering:** PostgreSQL, SQLAlchemy, Star Schema Design, ETL Pipelines, Data Modeling

ACADEMIC & LEARNING PROJECTS

Titanic Survival Prediction – Machine Learning Classification *[GitHub: github.com/KevinOcran/titanic]*

- Trained and compared 4 classification algorithms (Logistic Regression, Random Forest, XGBoost, Decision Tree) on Titanic dataset with 891 passengers
- Engineered features through one-hot encoding (categorical variables), log transformations (fare), median/mode imputation (missing values), and standardization (numeric features)
- Optimized Random Forest model using RandomizedSearchCV hyperparameter tuning, improving accuracy from baseline 62% to 83% on test set
- Evaluated models using multiple metrics (accuracy, precision, recall, F1-score) and created visualizations including confusion matrices, ROC curves, and feature importance plots

Online Retail Data Transformation – ETL Pipeline & Database Design *[GitHub: github.com/KevinOcran/online_retail]*

- Built ETL pipeline in Python to clean 541,909 retail transaction records containing duplicates, null values, and cancelled orders
- Designed star-schema database with fact table (transactions) and dimension tables (customers, products, dates) in PostgreSQL for analytics queries
- Implemented systematic data quality rules removing invalid records (negative quantities, missing customer IDs), improving clean data from ~70% to 95%+
- Used SQLAlchemy ORM to load cleaned data into PostgreSQL, enabling efficient SQL queries for business metrics (revenue by country, top products, customer segments)

GitHub Repository Activity Analysis – API Integration & Time Series Analysis *[GitHub: github.com/KevinOcran/openSourceHealthAnalysis]*

- Automated data collection from GitHub REST API with rate limit handling to extract commit history, contributor activity, and repository metadata
- Cleaned and structured nested JSON API responses into pandas DataFrames for time-series analysis of contribution patterns
- Created visualizations (heatmaps, line charts, correlation matrices) revealing weekly activity patterns and peak contribution times across multiple repositories

CERTIFICATIONS & TRAINING

- **Data Science and Analytics Certificate** – Thrive Africa (3-month intensive program, 2024)
- **Machine Learning and AI Certificate** – Thrive Africa (2024)
- **Generative AI and Advanced Prompt Engineering Certificate** – Thrive Africa (2024)
- **AWS Cloud Computing Essentials Certificate** – Thrive Africa (2024)
- **ChatGPT Prompt Engineering for Developers** – DeepLearning.AI & OpenAI (2024)

VOLUNTEER EXPERIENCE

Mathematics & Computing Instructor

Local Church School • 4 months • Primary 4 through JHS 3

- Taught mathematics fundamentals and basic computing concepts to students across 10 grade levels (ages 9-16)
- Created lesson plans and educational materials adapted to different learning abilities and age groups
- Developed communication and presentation skills while explaining technical and mathematical concepts to non-technical audiences

ADDITIONAL INFORMATION

- **Portfolio:** Complete project documentation and code samples available at kevinocran.github.io
- **Career Goal:** Seeking 3-6 month data science internship to apply academic knowledge, develop professional skills, and contribute to real-world data projects
- **Interests:** Calisthenics, anime, poetry, music curation, continuous learning in data science and machine learning